

**STI Ormoc Smart Library Management System with Digital Cataloging,
Automated Circulation, Inventory Management, QR-Based Attendance
Monitoring, and Online Printing Service Request**

**A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Ormoc**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

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May 21, 2026

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **STI Ormoc Smart Library Management System with Digital Cataloging, Automated Circulation, Inventory Management, QR-Based Attendance Monitoring, and Online Printing Service Request**

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

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MAY 21, 2026

APPROVAL SHEET

This capstone project proposal titled **STI Ormoc Smart Library Management System with Digital Cataloging, Automated Circulation, Inventory Management, QR-Based Attendance Monitoring, and Online Printing Service Request**, prepared and submitted by **Buentheo Nathaniel A. Noval, John Bazty C. Cantay, Keith Gaudi Gerard P. Juntilla**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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Accepted and approved by the Capstone Project Review Panel
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TABLE OF CONTENTS

	Page
Title Page	i
Endorsement form for Proposal Defense	ii
Approval Sheet	iii
Table of Contents	iv
Introduction	1
Project Context	1
Purpose and Description	2
Objectives	3
Scope and Limitations	4
Review of Related Literature/Studies/Systems	6
Methodology	10
Technical Background	10
Requirements Analysis	13
Requirements Documentation	14
Design of Software, System, Product, and/or Processes	17
References	43
Appendices	45
Resource Persons	45
Personal Technical Vitae	46
Transmittal Letter	51
Project Documentation	53

INTRODUCTION

Project Context

In the modern educational landscape, the library remains a central hub for academic activity; however, the global shift toward digitalization has redefined how students interact with information. Historically, libraries were managed through manual ledgers and index cards, but as technology evolved, the concept of a "Smart Library" emerged to address modern efficiency needs. Despite these trends, many regional institutions in the Philippines, including STI College Ormoc, continue to operate with manual systems that fail to meet the fast-paced demands of 21st-century learners.

The prevailing problem situation at STI College Ormoc is characterized by a heavy reliance on manual labor. Currently, automated circulation is non-existent. The "borrow and return" cycle requires students to manually sign index cards, while librarians must hand-write transaction details. This approach makes inventory management a significant challenge, as tracking the physical condition and location of resources requires constant manual auditing. Monitoring overdue materials presents an additional challenge. In the absence of automated notifications, librarians must manually review index cards to identify outstanding loans. This delayed tracking complicates penalty assessments and leads to missing resources, which leaves students without essential learning materials.

Handling printing manually makes it harder for the library to run smoothly, often causing crowded desks and slow service. Students must physically go to the library just to submit a file for printing, it creates unnecessary crowds and wastes a lot of time. This manual printing workflow requires students to submit their files physically via USB, Email, or Messenger, after which the librarian must manually check file formats and page counts, set print options, calculate costs, collect cash payments, monitor the queue, check the printed output, and physically record the transaction details in a traditional paper logbook.

The current manual log-in process forces users to wait in lines to fill out physical paper attendance forms. Users must handwrite details including the current date, time-in, full name, educational level (Senior High School or College), course/strand, section, and their specific reason for visiting (Library Visit, Printing, or Photocopy). This is followed by a mandatory signature and physical verification by the librarian before the student is allowed to enter the library premises.

This project is justified by the need to bridge the gap between traditional manual resources and modern digital expectations. By addressing these unresolved elements, the library can transition from a place of paperwork to a more efficient study environment. Therefore, there is a clear relationship between the need for operational modernization and the development of a localized smart management system.

Purpose and Description

The primary purpose of the STI Ormoc Smart Library Management System is to modernize and centralize the various services offered by the school's library into a single, user-friendly digital platform. The system functions as an all-in-one assistant for both students and library staff, moving away from the slow and outdated manual methods of the past. Its main goal is to make finding, borrowing, and returning books as easy as possible, while also introducing new digital conveniences that save time for everyone involved. By automating the repetitive tasks that currently take up much of the librarian's day, the system allows the library to focus more on its role as a helpful study environment rather than a place for paperwork.

What makes this project unique and innovative is how it tackles the specific, real-world problems faced by students on a daily basis. Beyond just tracking books, the system introduces a smart printing request feature. This allows students to handle their printing needs from their own devices instead of having to manually go to the library and hand

over the files via Gmail or USB. Additionally, the project is relevant because it respects the physical space of the library. By utilizing QR codes for check-in, the system offers a seamless experience and ensures precise attendance logging. This combination of traditional library tools and modern digital services makes the system a truly "smart" solution tailored to the actual needs of the campus community.

The project is also designed to be highly engaging and helpful for the modern learner. It includes a personalized dashboard where students can see exactly what they have borrowed and when they need to return it, helping them stay organized and avoid penalties. Furthermore, it adds a touch of innovation through a recommendation feature that suggests books based on what is popular or relevant to a student's specific course. By making the library's inventory more visible and accessible through a digital catalog, the system ensures that the school's resources are used to their full potential. This project is a relevant step forward, turning a traditional facility into a modern, efficient, and student-centered academic hub.

Objectives

General Objective

To design, develop, and implement the STI Ormoc Smart Library Management System that automates traditional library operations, including circulation, inventory tracking, and student services, to enhance the overall academic experience at STI College Ormoc.

Specific Objectives

- To develop a Digital Cataloging interface that enables users to search for books and research papers with accurate, up-to-date time availability status.
- To create an Automated Circulation module using QR or barcode scanning to facilitate faster check-in and check-out processes for library materials.
- To build an Inventory Management system that allows librarians to monitor the

physical condition of resources and generate automated inventory reports.

- To implement a QR-based attendance monitoring feature that tracks student entry and exit and identify peak usage times.
- To integrate an Online Printing Service Request system allowing students to upload files and schedule pickup times to reduce physical queues and the need for in-person desk visits.
- To provide an Admin Analytics Dashboard for library staff to view insights on the most borrowed resources and student usage trends.
- To develop a Fines and Accounts Clearance module that automatically computes overdue penalties based on library operating hours, monitors outstanding balances, and dynamically updates student clearance standing.
- To build a Printing Service Management and Consumables Tracking subsystem for library administrators to manage printing queues, monitor revenue, and track real-time physical supplies such as ink levels and bond paper stock.
- To evaluate the system's performance and user satisfaction based on the feedback of students and faculty members at STI College Ormoc.

Scope and Limitations

Scope

The project focuses on the development of a dual-platform system comprising a web-based application for administrative management and a mobile-responsive web platform for student access. The primary users are the students and faculty members of STI College Ormoc, who will utilize the digital catalog and printing services, and the library staff, who will manage the inventory and circulation. This research and development project is conducted during the second semester of the 2025–2026 academic year.

- Digital Cataloging: A searchable database for books and research papers with real-

time availability status.

- Automated Circulation: A QR-based system for the efficient borrowing and returning of library materials.
- Inventory Management: Automated tracking of book conditions, new arrivals, and stock levels for administrative use.
- QR-Based Attendance Monitoring: A module to track library entry and exit to monitor occupancy levels.
- Online Printing Service Request: A web-based feature allowing students to upload files and schedule pickup times for printed documents.
- Admin Analytics: A dashboard providing insights into peak usage hours and popular library resources.
- Reports Generation: An administrative module capable of generating exportable and printable reports on inventory audits, daily attendance statistics, printing transactions, and outstanding student penalty fines for unreturned books.

Limitations

- The system is specifically designed for the library of STI College Ormoc and may not be fully compatible with the infrastructure of other STI campuses without modification.
- The system requires a continuous internet connection to operate, which may limit its accessibility in areas with inconsistent network coverage.
- The Online Printing Service Request feature only manages file submission and scheduling of pickup times and does not include online payment processing or automatic printer maintenance functions.
- The project only covers library-related services and does not integrate with other institutional systems such as accounting, registrar, or learning management systems.

Review of Related Literature/Studies/Systems

The rapid digital transformation of academic libraries has led to the development of smart library management systems that integrate cataloging, circulation, user services, and analytics into unified platforms. Recent works on web-based and intelligent library management systems emphasize replacing manual, paper-based processes with automated modules for book transactions, member registration, and inventory control to improve accuracy and efficiency in service delivery (Chen & Ahmed, 2024; Kumar, 2023). In line with this shift, digital cataloging, automated circulation, attendance monitoring, and online service requests have become central themes in contemporary library technology research, providing the conceptual and technical foundation for the proposed STI Ormoc Smart Library Management System.

Local Institutional System Developments

A relevant local study for this project is the STI College Ormoc Online Library System with Electronic Access for Reservation and SMS Notification, developed by Gormano and Romo (2017). Their system was built using a WAMP server with a PHP backend and used JavaScript and AJAX to keep the user interface smooth. It aimed to move the library away from paper logbooks by digitizing basic records (like book profiles, categories, and borrower details), tracking daily transactions (borrowing, returning, and reserving), and sending out automated text reminders. While their project proved that digital reservations and notifications could work on campus, it only handled basic circulation and relied on extra SMS hardware to send texts. The proposed STI Ormoc Smart Library Management System builds directly on what they started but upgrades it into a dual-platform web and mobile-responsive system. By adding practical, low-cost features that the 2017 version lacked, such as QR codes for quick borrowing and attendance, an online printing queue, live analytics dashboards, and downloadable admin reports, this new project fills the remaining gaps to create a truly modern study environment for the school.

Digital cataloging and OPAC services

Digital cataloging techniques have evolved from physical index cards to metadata-driven, searchable databases that provide real-time resource availability (Kumar, 2023; Singh & Patel, 2022). Adopting these systems significantly enhances student information-seeking behavior by offering comprehensive keyword searching and live status updates (Rahman & Lee, 2025). While existing research often focuses on complex AI-assisted cataloging for large-scale universities, a gap remains for smaller institutions. The proposed STI Ormoc system addresses this by prioritizing a user friendly interface and real-time availability, avoiding the resource-heavy overhead associated with high-end AI implementations (Chen & Ahmed, 2024).

Automated circulation and transaction management

Automated circulation modules are critical for modernizing library transactions, handling everything from user registration and book loans to overdue notifications in a unified workflow (Gupta, 2020; National Library Network, 2019). Research indicates that these systems significantly improve transaction speed, record accuracy, and user satisfaction compared to manual methods (Iqbal & Mahmood, 2021). While RFID-based systems offer advanced tracking, they often require expensive hardware incompatible with smaller campus budgets (Delos Santos & Cruz, 2022). The STI Ormoc system leverages existing camera and mobile technology to achieve these same benefits in speed and efficiency without the prohibitive costs of specialized RFID infrastructure (Lopez & Chandra, 2025).

Inventory management and analytics dashboards

Integrated inventory management systems allow for real-time tracking of collection health and acquisition history, effectively removing the manual labor of stock auditing (Kumar, 2023). When linked with administrative analytics dashboards, these modules

enable librarians to monitor usage patterns, such as peak circulation hours and popular resources, transforming library management from reactive to data-driven (Iqbal & Mahmood, 2021; Lopez & Chandra, 2025). The STI Ormoc system adopts this approach by providing a dashboard that informs collection development and student recommendations, ensuring library resources are utilized to their full potential based on actual campus demand (Singh & Patel, 2022).

QR-based attendance monitoring

QR code-based attendance systems have emerged as a highly effective, low-cost solution for tracking library occupancy and student flow (Putra et al., 2025). Systematic reviews show these systems reduce administrative time, minimize human error, and provide reliable digital logs for planning and security (Putra et al., 2025; Olanrewaju & Bamidele, 2024). By implementing a QR-based attendance module at STI Ormoc, the library can provide real-time occupancy data to students while creating an automated history of usage, offering a cost-effective alternative to expensive sensor-based or RFID hardware (Delos Santos & Cruz, 2022; Reyes, 2016).

Online printing service requests

The growth of virtual library services includes the rise of online platforms for supporting student needs, such as off-site printing and copying. Research highlights that moving these requests to a digital platform decentralizes service, clears physical congestion at library counters, and increases operational efficiency (Rahman & Lee, 2025). By embedding an online printing request feature directly into the STI Ormoc Smart Library Management System, the project bridges the gap between digital resource search and physical library services, creating a seamless, self-service experience that accommodates modern student schedules (Olanrewaju & Bamidele, 2024; Rahman & Lee, 2025).

Synthesis

The reviewed literature reveals a significant shift in academic libraries from fragmented, manual record-keeping to integrated, data-driven digital management, demonstrating that automating core operations like cataloging, circulation, and inventory is essential for modernizing library services and enhancing user accessibility. Research consistently shows that integrating emerging technologies, specifically QR-based attendance monitoring and web-based request platforms, provides low-cost, scalable solutions that decentralize services and optimize space, effectively meeting the needs of modern, mobile-dependent students. The proposed STI Ormoc Smart Library Management System aligns with this research trajectory; by unifying these disparate functions into a single platform, it addresses the identified gap in the literature regarding efficient, budget-friendly digital solutions for smaller academic institutions. Ultimately, this project replaces labor-intensive manual processes with a streamlined digital framework, supporting both the administrative objectives of the STI library and the academic requirements of the STI community through evidence-based, smart technology integration.

METHODOLOGY

Technical Background

- **Technologies to be Used**

To develop and implement the Smart Library Management System, the researchers will utilize a modern tech stack focused on high performance and cross-platform accessibility. The frontend will prioritize user experience through a web application built with HTML, Tailwind CSS, and JavaScript, leveraging React.js to create a highly dynamic and interactive interface. To meet the growing demand for mobile-first solutions, the system will also include a React Native mobile application, providing students and faculty with seamless, cross-platform experience for searching catalogs and managing reservations directly from their mobile devices.

The system's engine and data integrity are powered by a robust backend and secure storage architecture. Node.js with Express.js will handle the core business logic, managing real-time requests for transactions like book borrowing and printing services. Data will be structured and stored using MySQL, ensuring that sensitive information, including user accounts, inventory records, and borrowing histories, is organized and reliable. To safeguard this data, the researchers will implement rigorous security measures such as password encryption, secure authentication, and Role-Based Access Control (RBAC) to distinguish between administrative functions for librarians and standard access for general users.

Finally, the system integrates automated technologies and cloud infrastructure to ensure operational efficiency and data-driven insights. By utilizing barcode and QR code technologies via tools like html5-qrcode, the system automates attendance and inventory tracking, significantly reducing manual entry errors. The entire platform will be deployed

via cloud-based services like Vercel or Render for high availability and remote accessibility. To assist in administrative decision-making, the system will feature analytics dashboards built with Chart.js or Recharts, transforming raw data into visual reports on inventory trends, attendance, and printing statistics.

- **Calendar of Activities**

MONTH	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER
ACTIVITY												
Topic Selection & Proposal Planning												
Requirements Gathering												
System Analysis and Planning												
Proposal Writing & Documentation												
Proposal Presentation and Defense												
Design of UI & System Flow												
Database Design & Architecture												
Frontend & Mobile Development												
Backend & API Development												
Module Integration (Barcode, QR, Printing)												
System Testing and Debugging												
User Acceptance Testing (UAT)												
Final Documentation Completion												
Final Revision and Improvement												
Final Presentation and Defense												

Ethan Noval

Bazly Cantay

Keith Juntilla

All Members



- **Resources**

To successfully develop, test, and implement the Smart Library Management System, the following hardware and software resources are required:

Hardware Resources

- **Development Computers:** High-performance laptops or desktop computers (minimum Core i5 processor and 8GB RAM) capable of running integrated development environments (IDEs) and local servers.
- **Mobile Devices:** Smartphones used to test the responsiveness and functionality of the React Native mobile application.

- **Barcode and QR Scanners:** Dedicated USB barcode scanners or high-resolution webcams used to facilitate and test the automated book-tracking and attendance features.
- **Network Equipment:** High-speed internet routers and connection devices to support cloud deployment, database synchronization, and real-time notifications.

Software Resources

- **Development Environment:** Visual Studio Code (VS Code) as the primary code editor, along with the Node.js runtime environment for backend execution.
- **Database Management:** MySQL for structured data storage and MySQL Workbench for designing and managing database schemas.
- **Design and Prototyping Tools:** Figma or Canva for creating the User Interface (UI) mockups and mapping the User Experience (UX) flow.
- **Version Control:** Git and GitHub for source code management, collaborative tracking, and version history.
- **Deployment Platforms:** Vercel and Render for hosting the web application and backend services in a live environment.
- **Documentation Tools:** Microsoft Office Suite or Google Workspace for drafting the capstone manuscript, technical manuals, and project reports.
- **Web Browsers:** Modern browsers like Google Chrome or Microsoft Edge for debugging frontend components and testing cross-browser compatibility.

Requirements Analysis

The Requirements Analysis for the Smart Library Management System for STI Ormoc identifies the necessary computing solutions to address the specific needs of the institution by evaluating the stakeholders, activities, and existing limitations of the current setup.

The people involved include the library staff and librarians who manage the collection and monitor analytics, as well as the students and faculty members who utilize the library for borrowing, research, and printing services. Additionally, a system administrator ensures the platform remains secure and operational. Regarding the business activity, the system focuses on automating book cataloging, borrowing and return cycles, QR-based attendance monitoring, inventory management and printing request management. These functions are designed to operate within the environment of the STI Ormoc Library premises, while also being accessible via mobile devices and web browsers to allow for remote inquiries and reservations.

The timing of these operations primarily occurs during the library's standard hours of 7:00 AM to 5:00 PM, Monday through Saturday, with the system providing real-time updates for due dates and availability. Currently, the procedures are performed through labor-intensive manual methods, such as writing in logbooks, searching through card catalogs, and using physical stamps for due dates. These traditional processes are often prone to human error and lack efficiency. By replacing these manual workflows with barcode scanning, digital inventory analytics, and automated notification features, the proposed system modernizes the library's operations, ensuring a more accurate and seamless experience for all users.

Requirements Documentation

Functional Requirements

The proposed system is divided into two primary interfaces, each equipped with specific features designed to streamline library operations:

1. User Side Features (Students and Faculty)

- **User Authentication:** Secure registration and login protocols with account verification to protect user data.
- **User Dashboard:** A personalized interface displaying currently borrowed books, active reservations, borrowing history, and notifications, alongside a library map for physical navigation.
- **Advanced Search and Browse:** A multi-tier catalog discovery engine allowing patrons to query standard literature and academic papers simultaneously by Title, Author, or ISBN. The interface must feature client-side filtering by Category, Author, Publication Year, and Availability Status. When an item card is selected, the system pulls deep cataloging metadata to display dynamic shelf location, real-time availability, and present an Automated Reference Generator allowing students to view and copy standard APA citations instantly.
- **Borrowing and Returning:** Digital borrow requests are facilitated by barcode scanning. The system enforces a strict 1-day borrowing period with a due time of 8:59 AM on the next operating day. The system handles automatic due date monitoring and computes late penalties based on library operating hours.
- **Borrowing Constraints and Cart Limits:** To ensure equitable resource distribution, the system applies variable cart constraints during the digital reservation and checkout process. Student accounts are subject to a strict borrowing limitation, restricting them

to a maximum of two (2) books simultaneously. Faculty and staff accounts are exempted from this restriction and are permitted to add multiple items to their digital cart for academic and instructional use.

- **Penalty Computation Rules:** The system automatically assesses an hourly fine of ₱2.00 per hour for materials returned after the cutoff time on the due date. If the material remains unreturned, a flat daily rate of ₱10.00 per day is applied starting the following day.
- **Reservation System:** The ability to reserve currently unavailable books and join a digital waiting list with automated availability notifications.
- **Printing Service Request:** A module for uploading files and selecting print configurations, including the number of copies, print type (Colored or Not), and optional notes. The system automatically calculates costs, places the job into the print queue, and allows users to track the real-time status of printing requests as they transition from Pending to Printing to Ready for Pickup before payment is settled in the library.
- **QR Attendance Monitoring:** A contactless entry system where users scan a QR code to log their attendance automatically.
- **Notification System:** Automated alerts for due dates, overdue penalties, reservation arrivals, and printing updates.
- **Digital Clearance Status:** A dedicated module where users can view their real-time library clearance standing (e.g., "Cleared" or "Not Cleared"). If a student is blocked, the system explicitly displays the specific unreturned books or outstanding penalty fines causing the hold.

2. Librarian and Admin Side Features

- **Dashboard and Analytics:** A centralized, high-level administrative interface tracking twelve (12) distinct Real-Time Key Performance Indicators (KPIs) upon successful authentication: Total Books, Active Borrowed Books, Most Borrowed Books, Active Users, Daily Attendance, Available Books, Overdue Books, Active Reservations, Total Penalties, Recent Activities, Book Availability Stats, and Returned Books.
- **Book and Research Management:** Full CRUD (Create, Read, Update, Delete) capabilities for books and thesis materials, including barcode assignment and category management.
 - **Category Management Sub-Table:** The category interface must feature an explicit data grid displaying Category ID, Category Name, Description, Number of Books currently tied to that category, and Date Added, alongside direct triggers for Edit Category and Delete Category actions.
 - **Research and Thesis Inventory Analytics:** The research repository module must feature a dedicated data-visualization component capable of rendering a Department Analysis (aggregating and grouping submitted manuscripts by their respective academic programs) and Year Trends (tracking and graphing the volume of institutional research submissions across consecutive graduation years)
- **Transaction Monitoring:** A centralized hub to track all borrowing cycles, approve requests, monitor overdue items, and export detailed transaction reports.
- **Fines Management Hub:** A dedicated administrative workspace providing segmented visibility into financial penalties. This module must display an overview breakdown distinguishing Paid Fines from Unpaid Fines, render a comprehensive Fines Table mapping Book Name, User Name, Student ID, Fine Amount, and Payment Status, and provide explicit actions to Edit Fine, Delete Fine, or execute an Export CSV command specifically for accounting audits

- **Inventory Management:** Automated inventory audits using barcode scanning, tracking of missing or damaged resources, and comprehensive inventory reporting.
- **Printing Service Management:** Tools to manage print queues, update job statuses, and track revenue and usage statistics.
- **Printing Supplies Asset Management:** A dedicated inventory and analytics layer tracks physical library printing consumables in real time. The ink repository submenu tracks unique Ink IDs, ink cartridge types, color variations, remaining fluid levels, and specific printer assignments, throwing low-ink triggers when thresholds are breached. The bond paper stock matrix monitors paper size dimensions (e.g., Short, A4, Long), remaining reams, replenishment histories, and expense trends to generate data-driven supply consumption reports for administrative budgeting
- **Attendance and User Management:** Real-time monitoring of QR logs, generation of attendance reports, and the ability to manage user roles or account statuses.
 - **User State Lifecycle Actions:** The administration table must provide explicit operational triggers to instantly Activate or Deactivate patron profiles, shifting their active account status in real time to restrict or grant library privileges
- **Automated Clearance Management:** A master dashboard that automatically flags students as "Cleared" or "Pending" based on their transaction history. It allows librarians to search the student master list, view individual clearance blocks, manually override statuses for special cases, and export bulk clearance reports for campus submission.

Design of Software, System, Product, and/or Processes

To clearly map the operational transition from legacy procedures to the proposed digital ecosystem, this section presents three architectural workflows. Figure 1 (Manual Attendance Process) establishes the 11-step paper logbook path currently required for facility entry. Figure 3 (Manual Process of Printing) chronicles the physical medium vulnerabilities inherent to legacy file handling and cash recording. Figure 3 (Design Process) charts the unified software system architecture, outlining student access privileges, search indexing logic, dynamic automated queues, and immediate notifications.

MANUAL PROCESSES

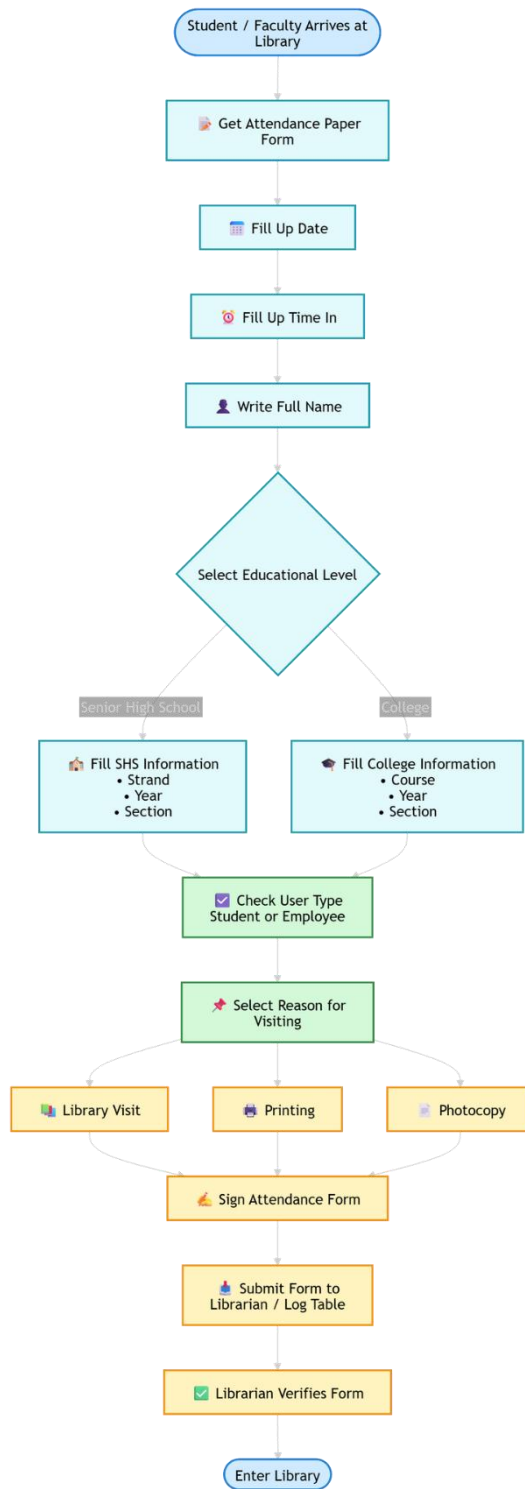


Figure 1. Manual Process Attendance

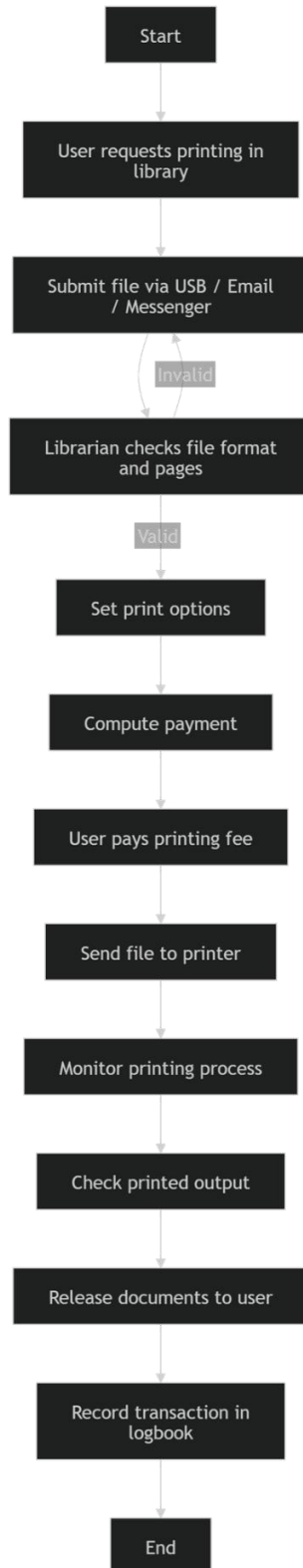


Figure 2. Manual Printing Process

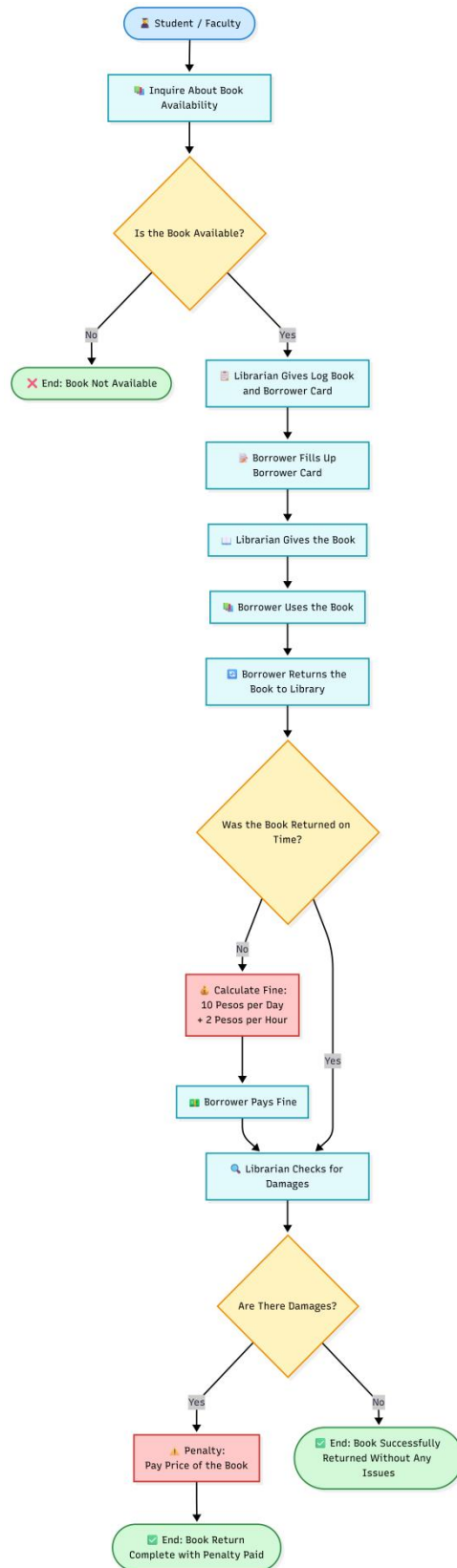


Figure 3. Manual Borrowing Process

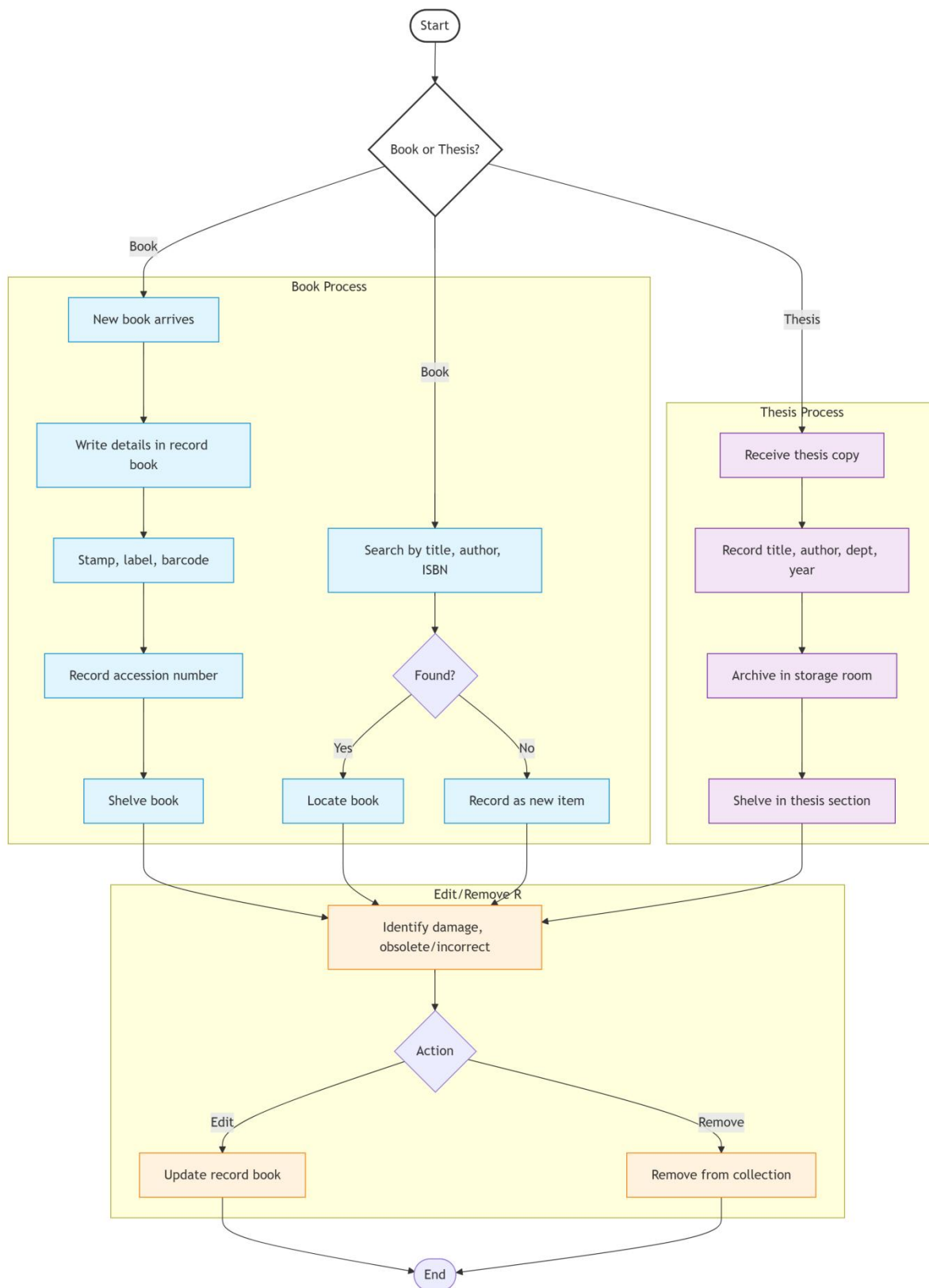


Figure 4. Manual Books Management Process

ADMIN SIDE

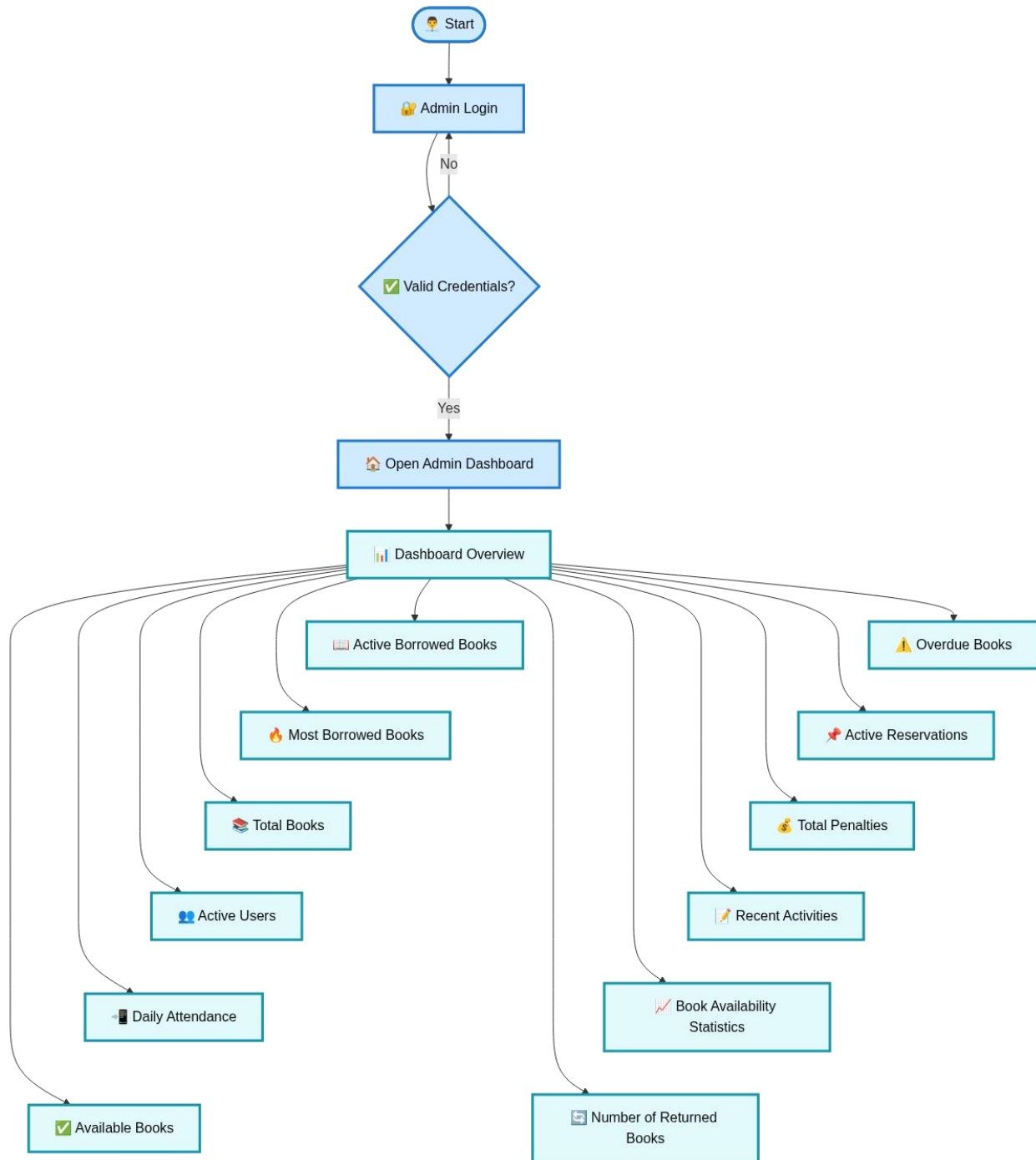


Figure 5. Proposed Process Dashboard (Admin Side)

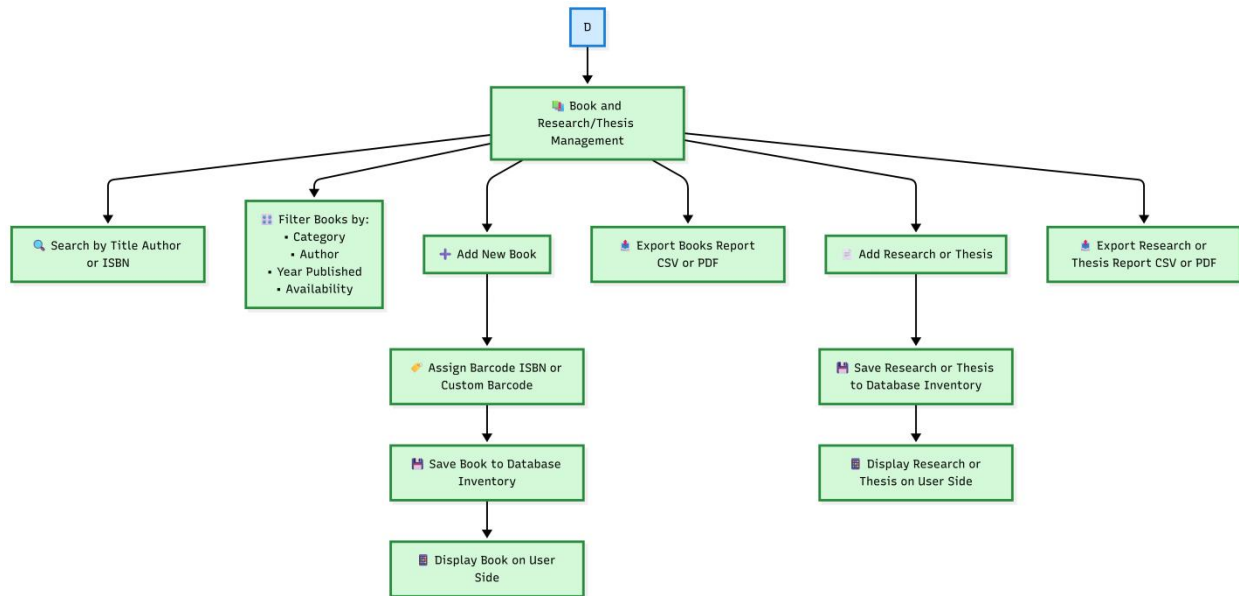


Figure 4. Proposed Book Management Process (Admin Side)

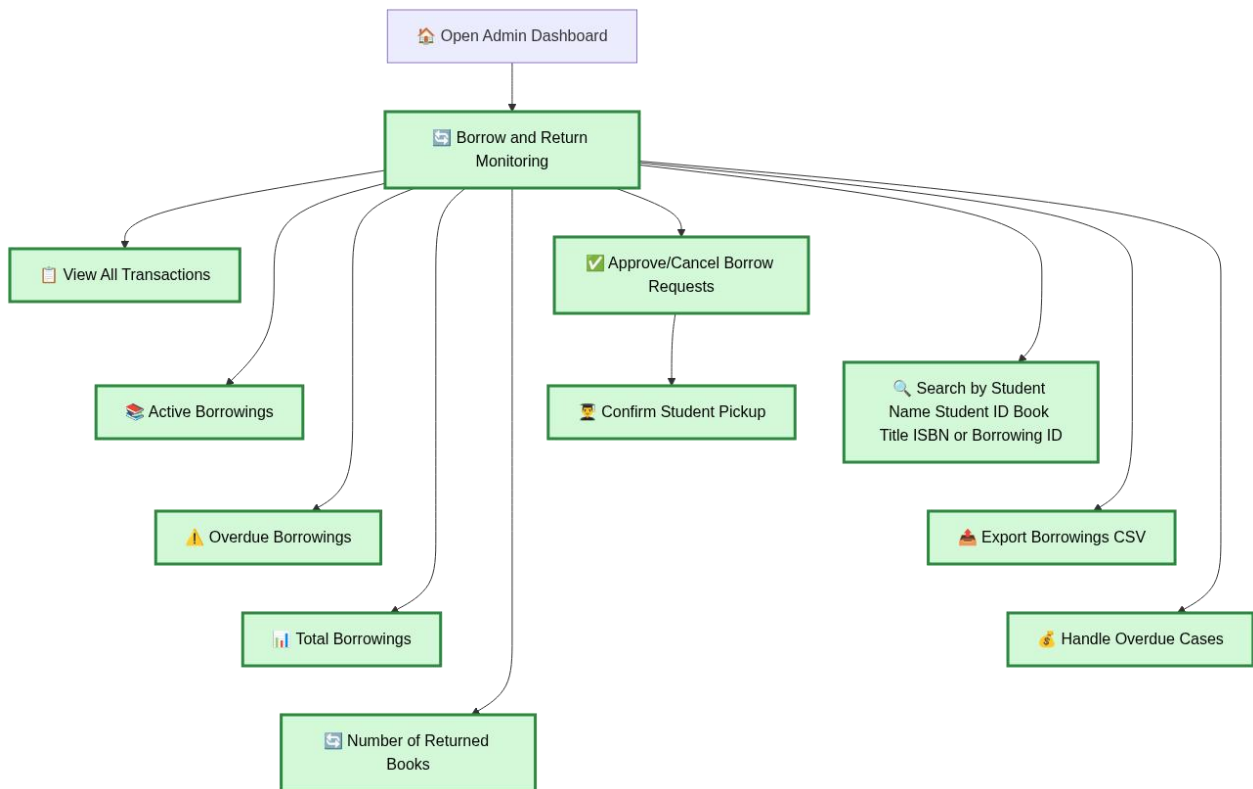


Figure 6. Proposed Borrow and Return Monitoring Process (Admin Side)

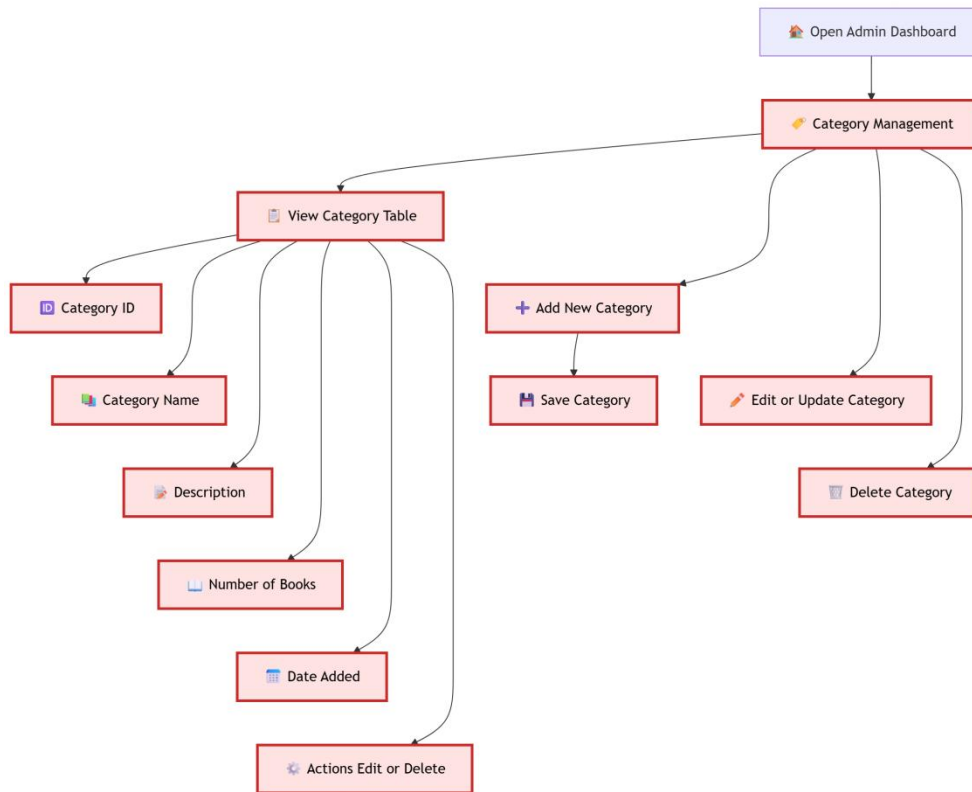


Figure 7. Proposed Category Management Process (Admin Side)

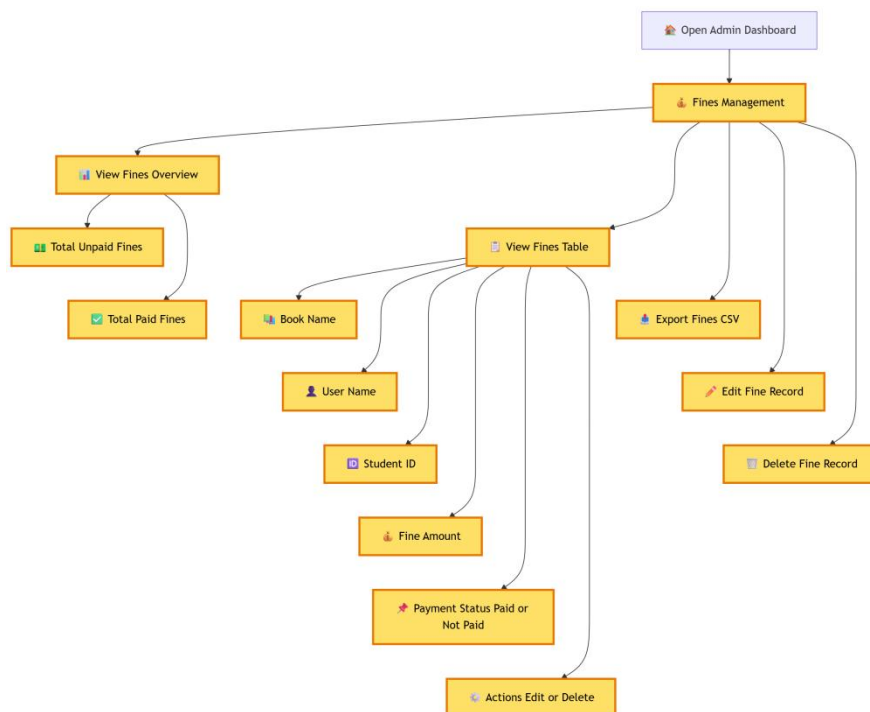


Figure 8. Proposed Fines Management Process (Admin Side)

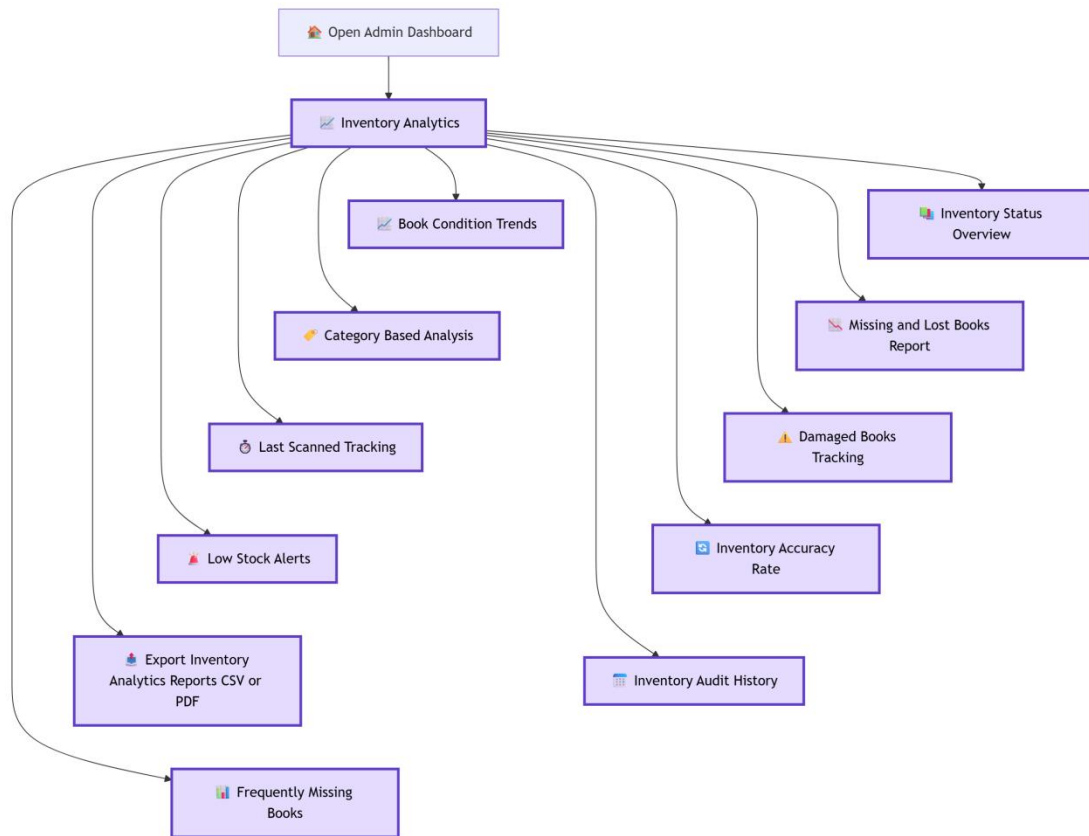


Figure 9. Proposed Inventory Analytics Process (Admin Side)

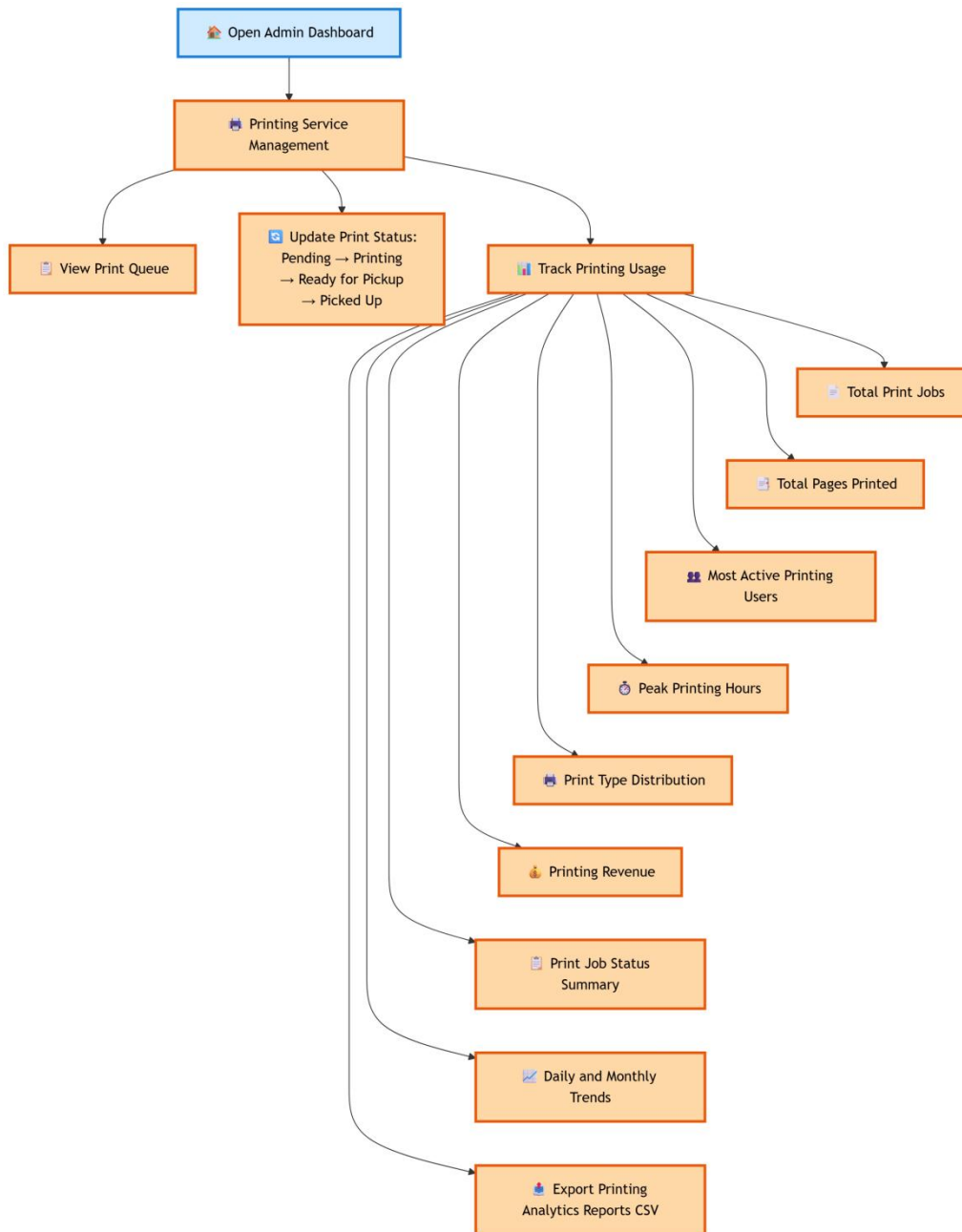


Figure 10. Proposed Printing Service Management Process (Admin Side)

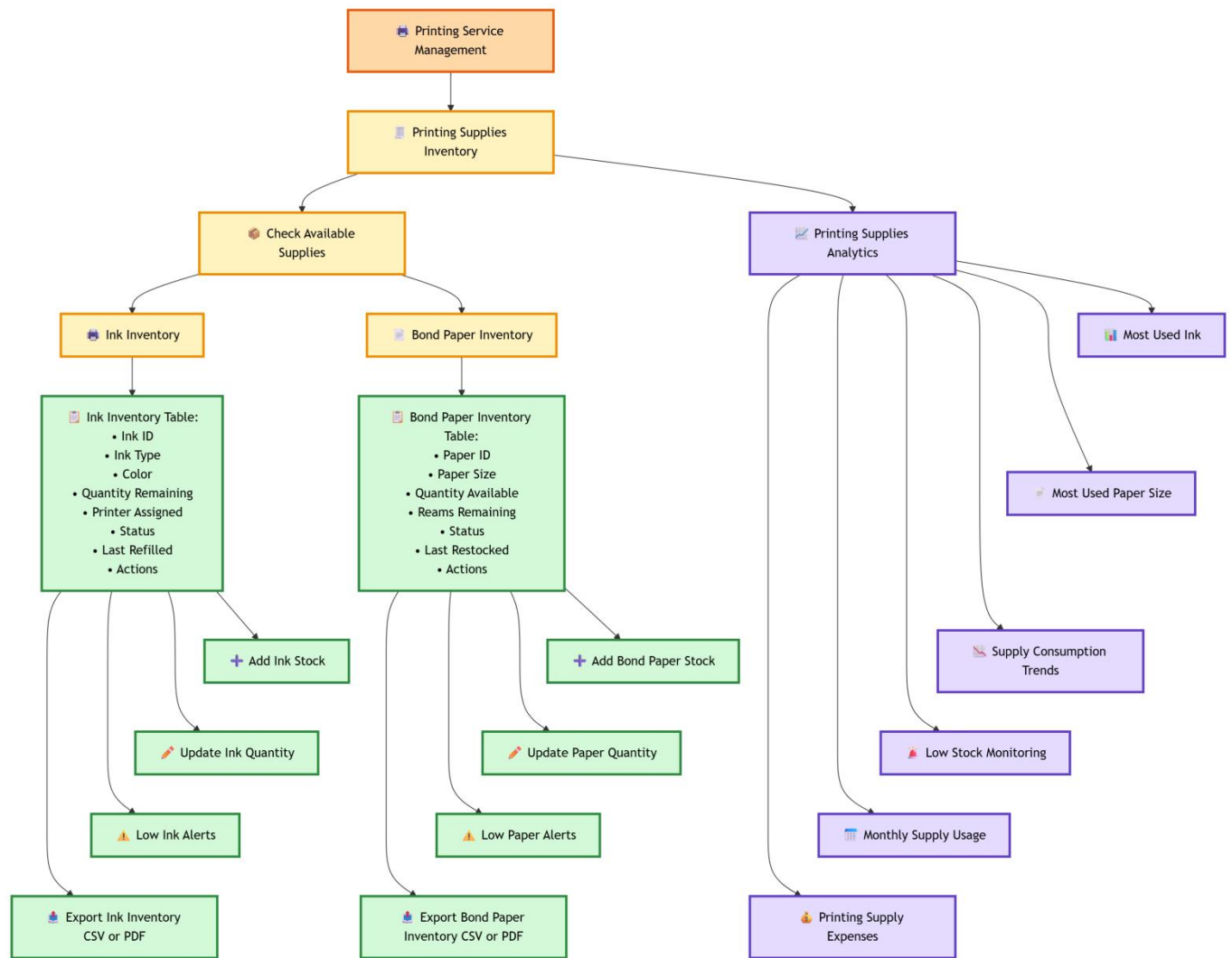


Figure 11. Proposed Printing Supplies Inventory/Analytics Process (Admin Side)

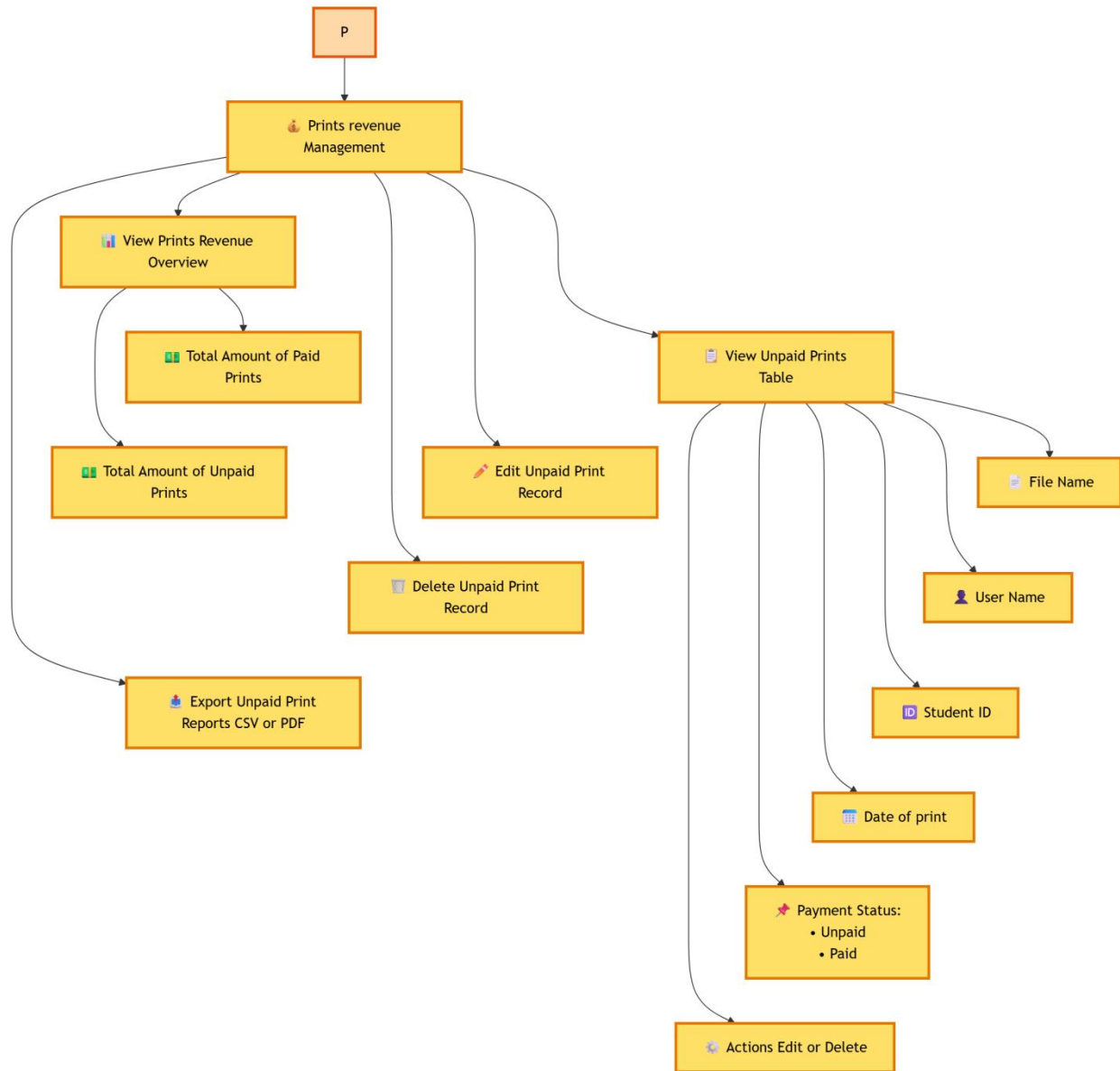


Figure 12. Proposed Prints Revenue Management Process (Admin Side)

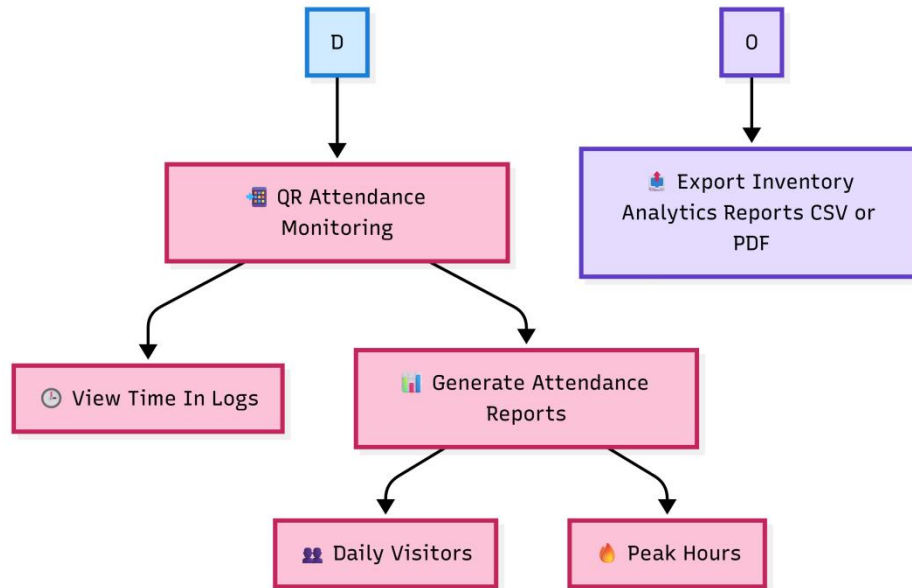


Figure 13. Proposed Attendance Monitoring Process (Admin Side)

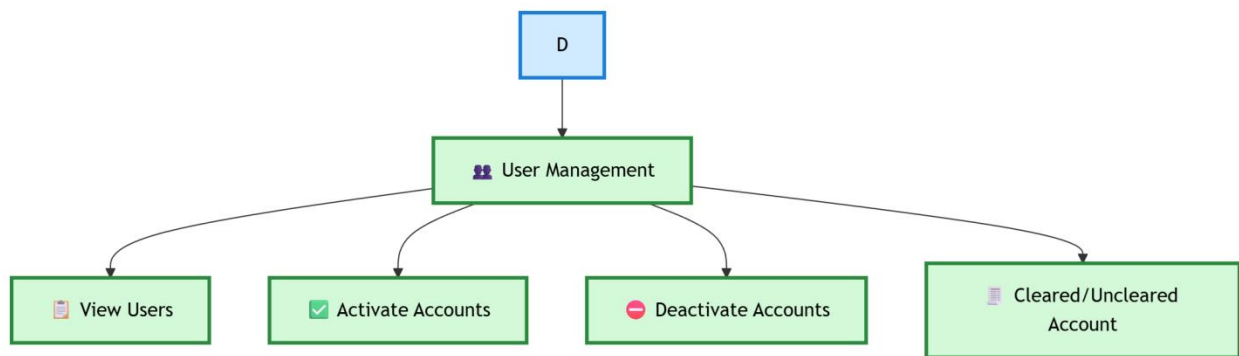


Figure 14. Proposed User Managment Process (Admin Side)

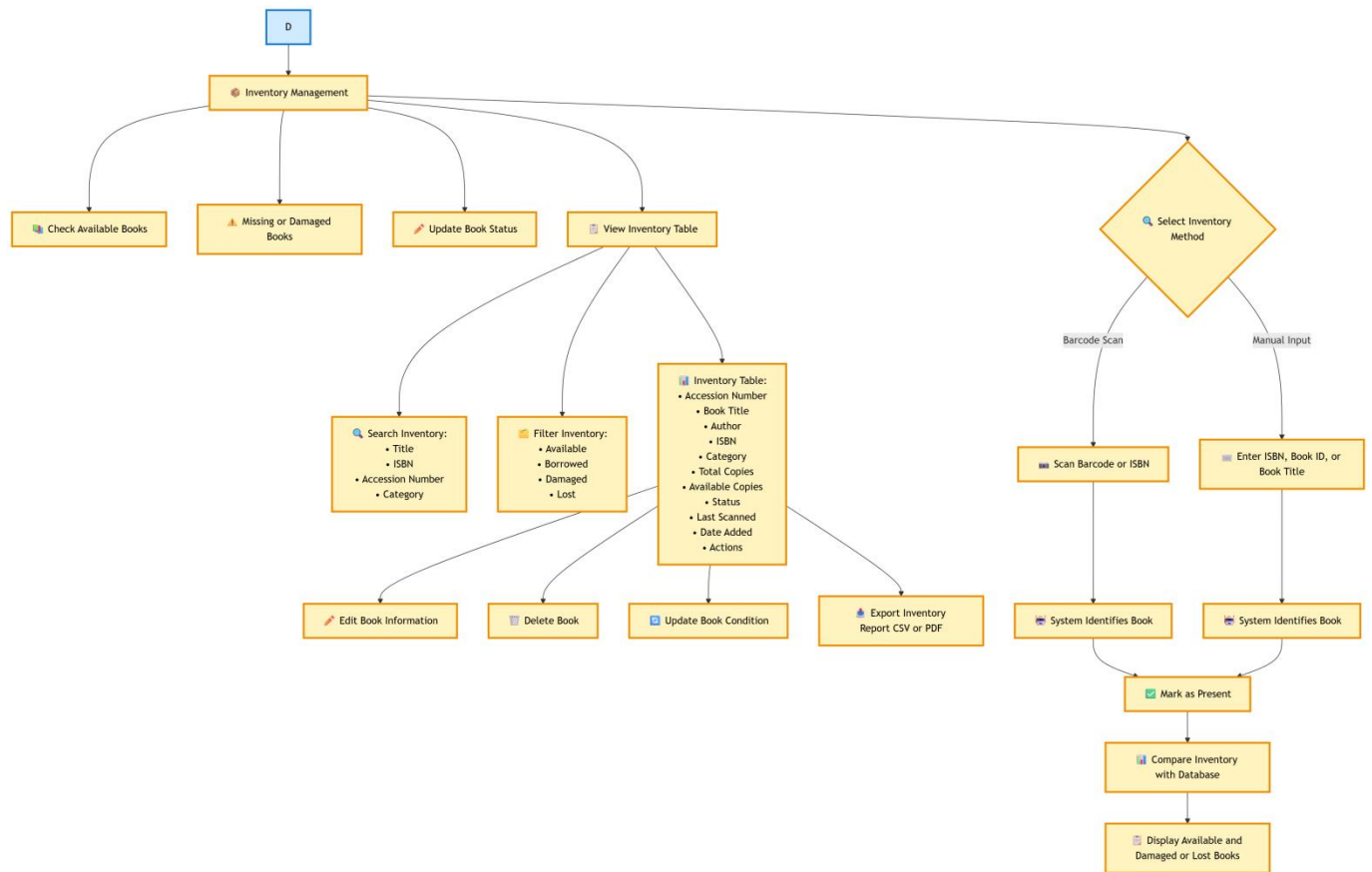


Figure 15. Proposed Inventory Management Process (Admin Side)

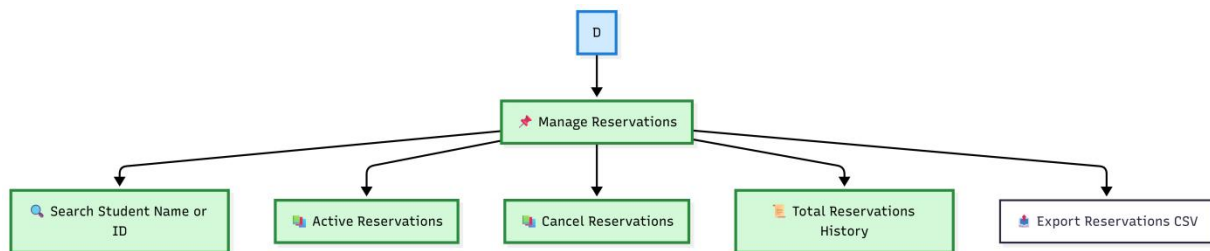


Figure 16. Proposed Reservation Management Process (Admin Side)

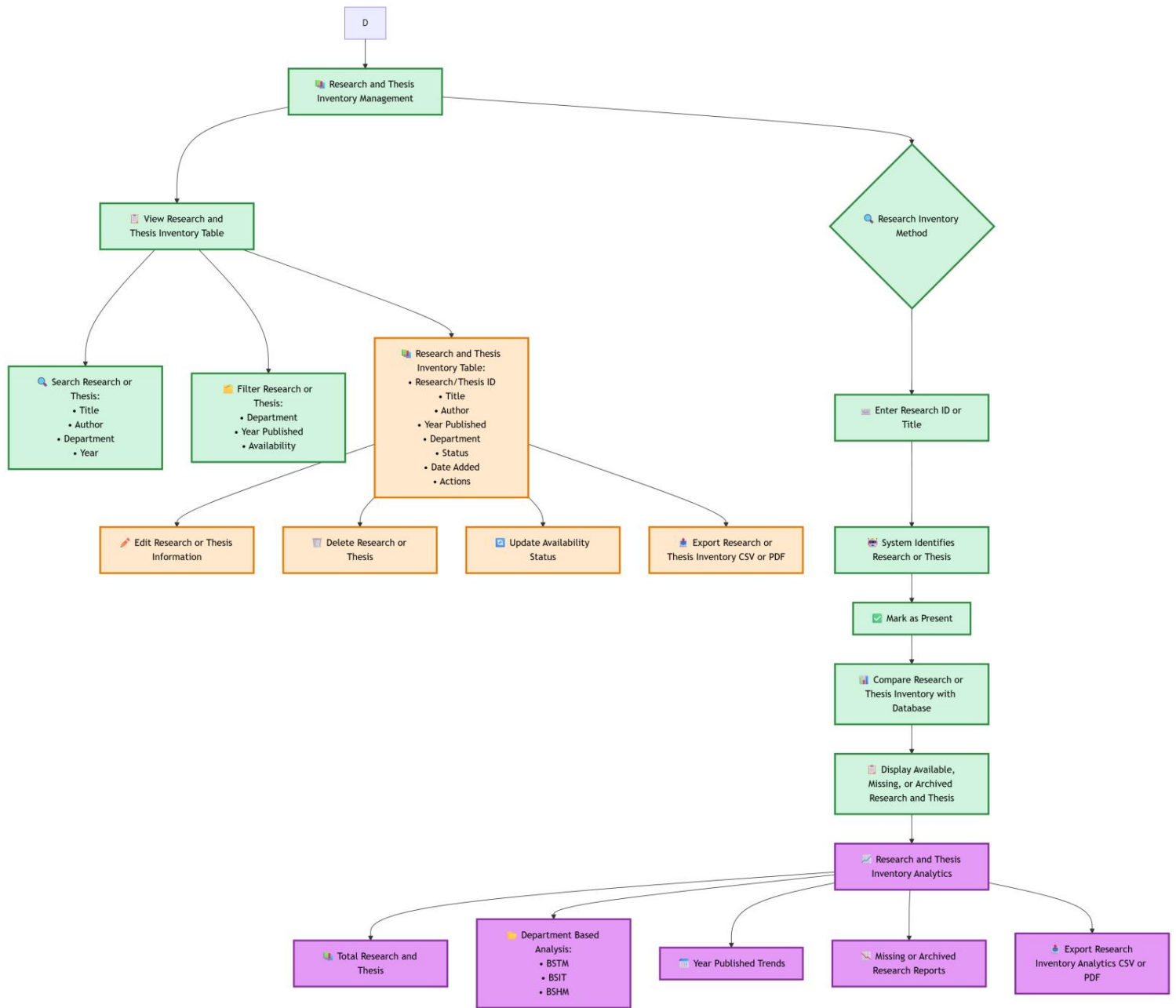


Figure 17. Proposed Research and Thesis Inventory Management Process (Admin Side)

USER SIDE

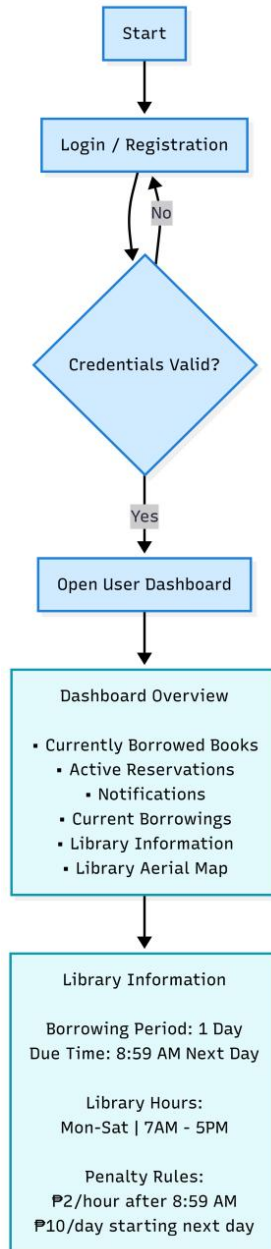


Figure 18. Proposed User Dashboard Process (User Side)

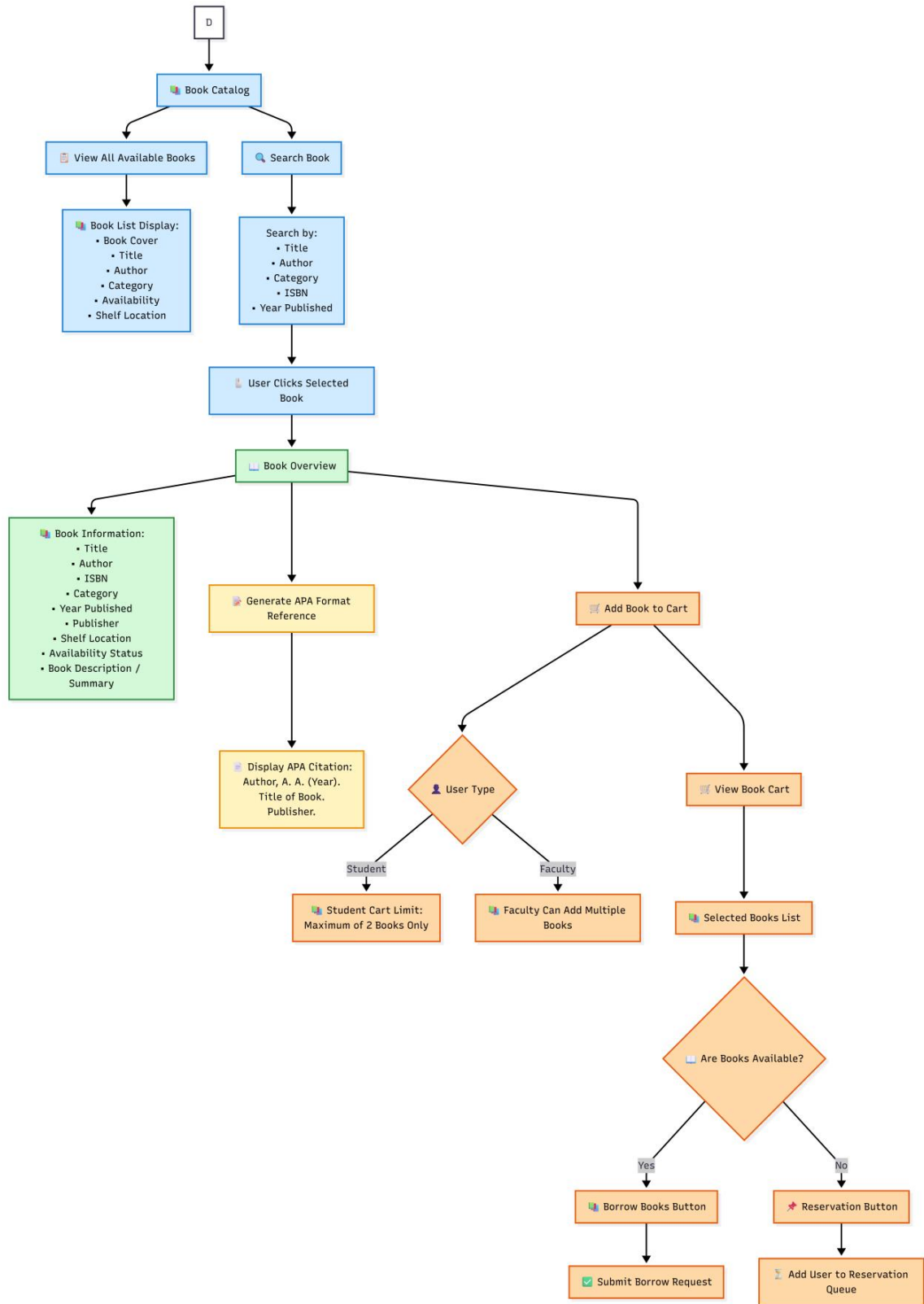


Figure 19. Proposed Book Catalog Process (User Side)

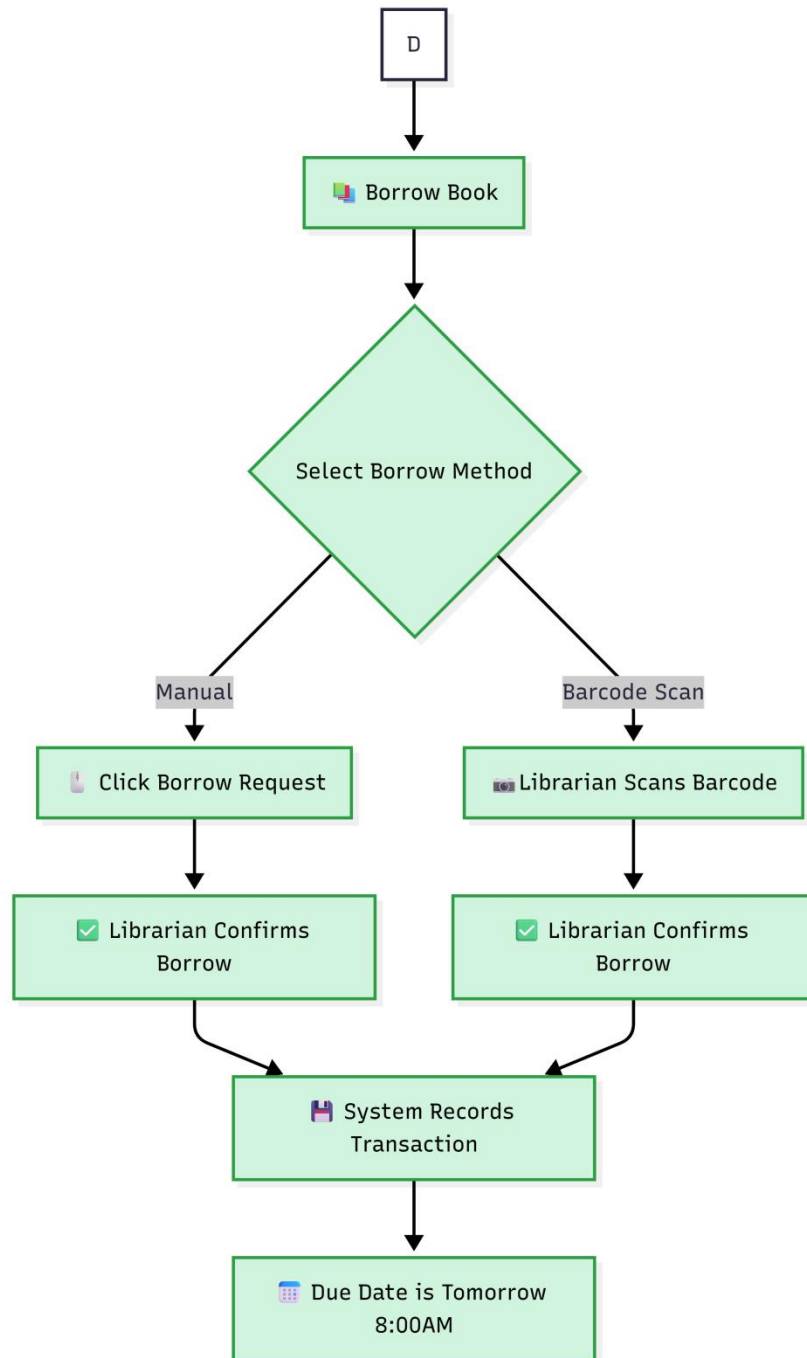


Figure 20. Proposed Borrowing Process (User Side)

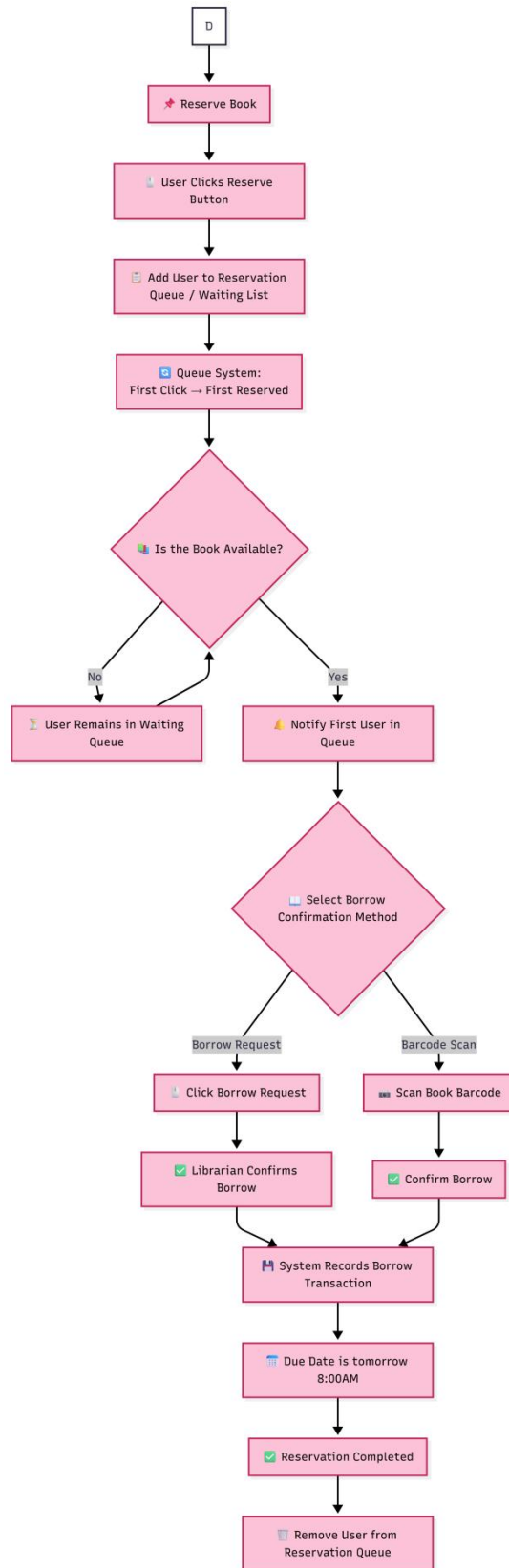


Figure 21. Proposed Reservation Process (User Side)

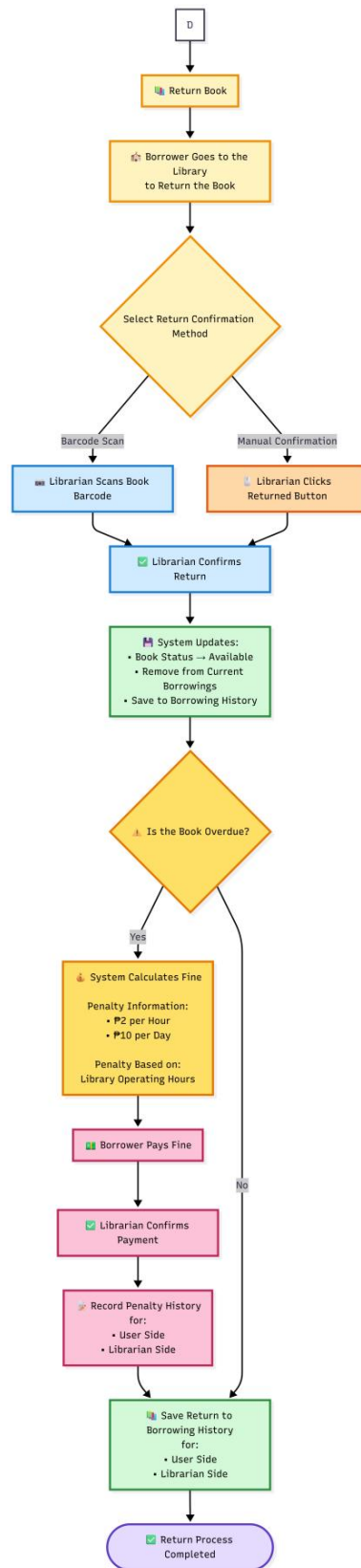


Figure 22. Proposed Returning Process (User Side)

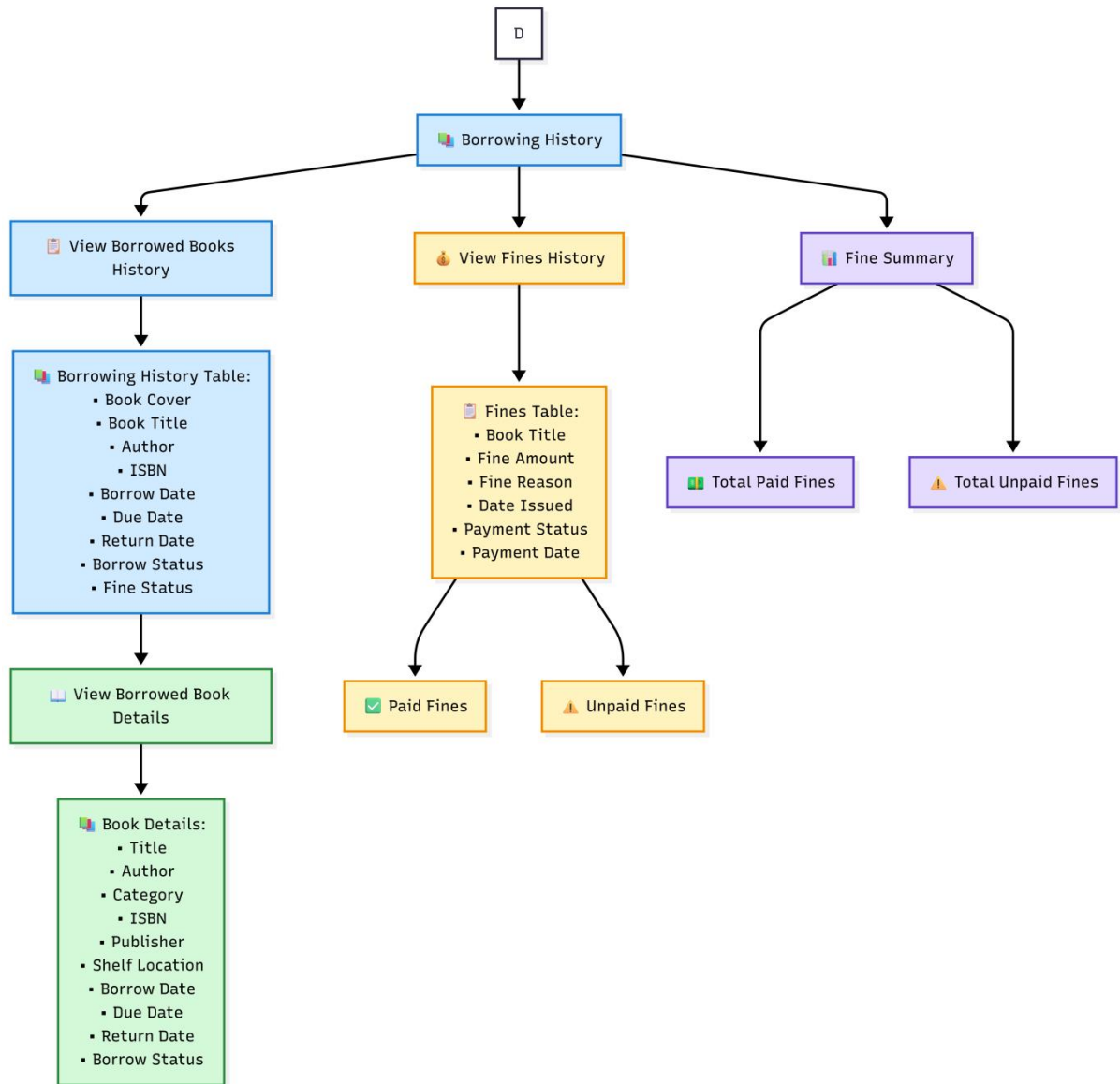


Figure 23. Proposed Borrowing History Process (User Side)

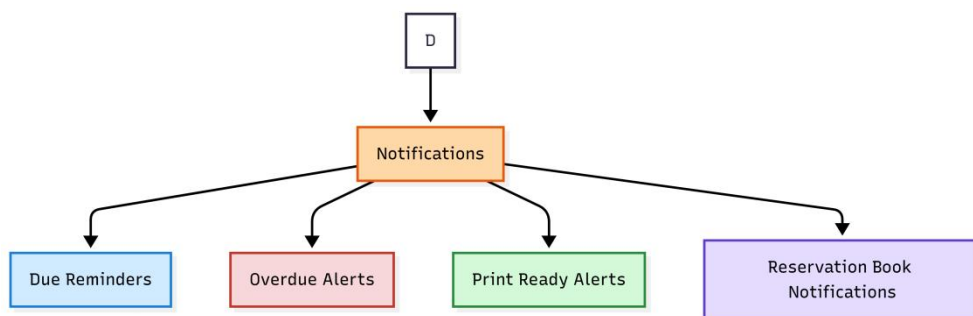


Figure 24. Proposed Notifications Process (User Side)

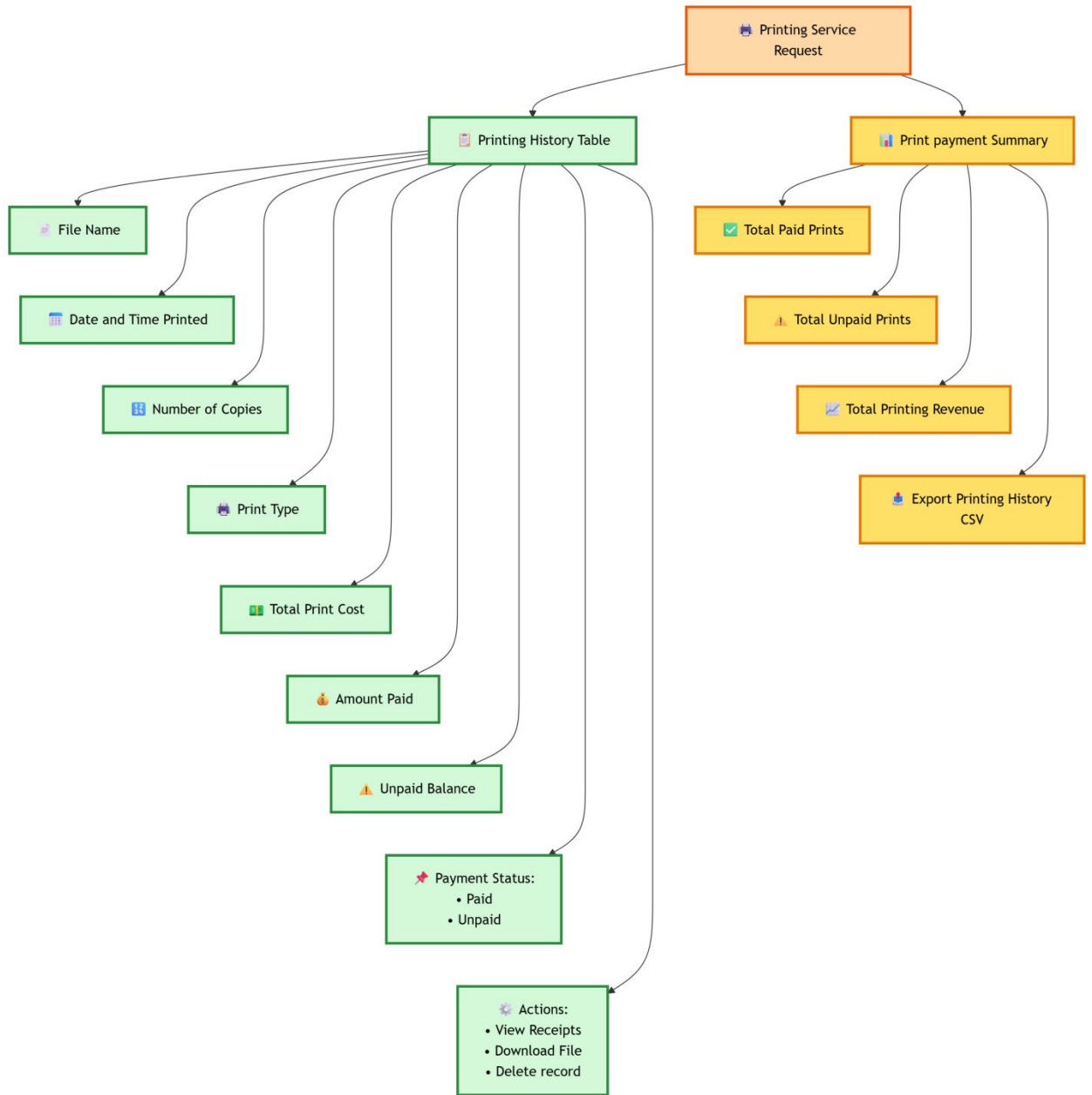


Figure 25. Proposed Print History and Print Payment Summary Process (User Side)

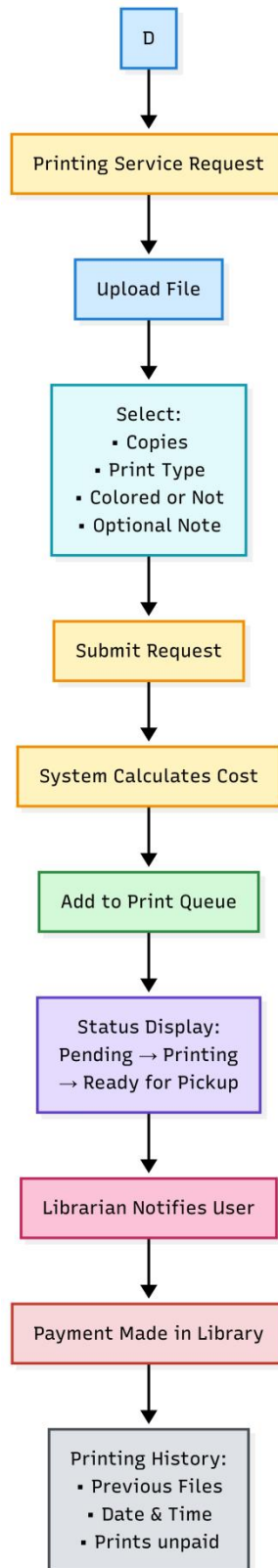


Figure 26. Proposed Printing Service Request Process (User Side)

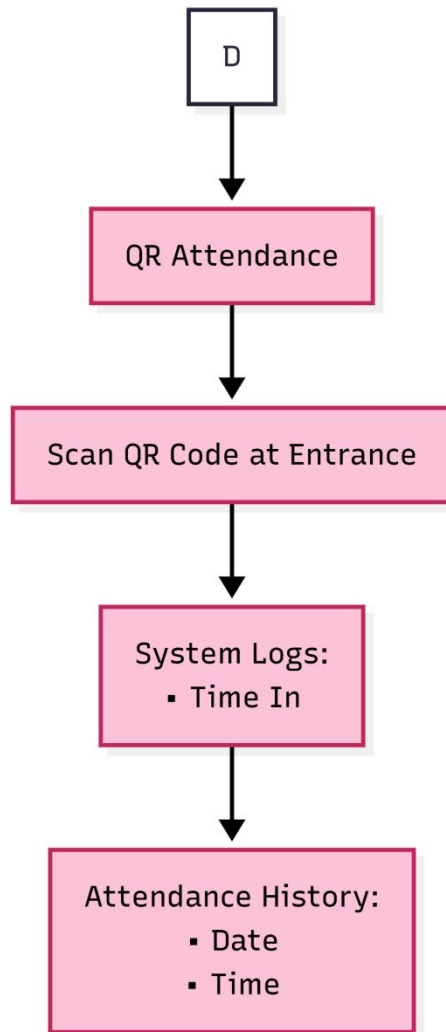


Figure 27. Proposed QR Attendance Process (User Side)

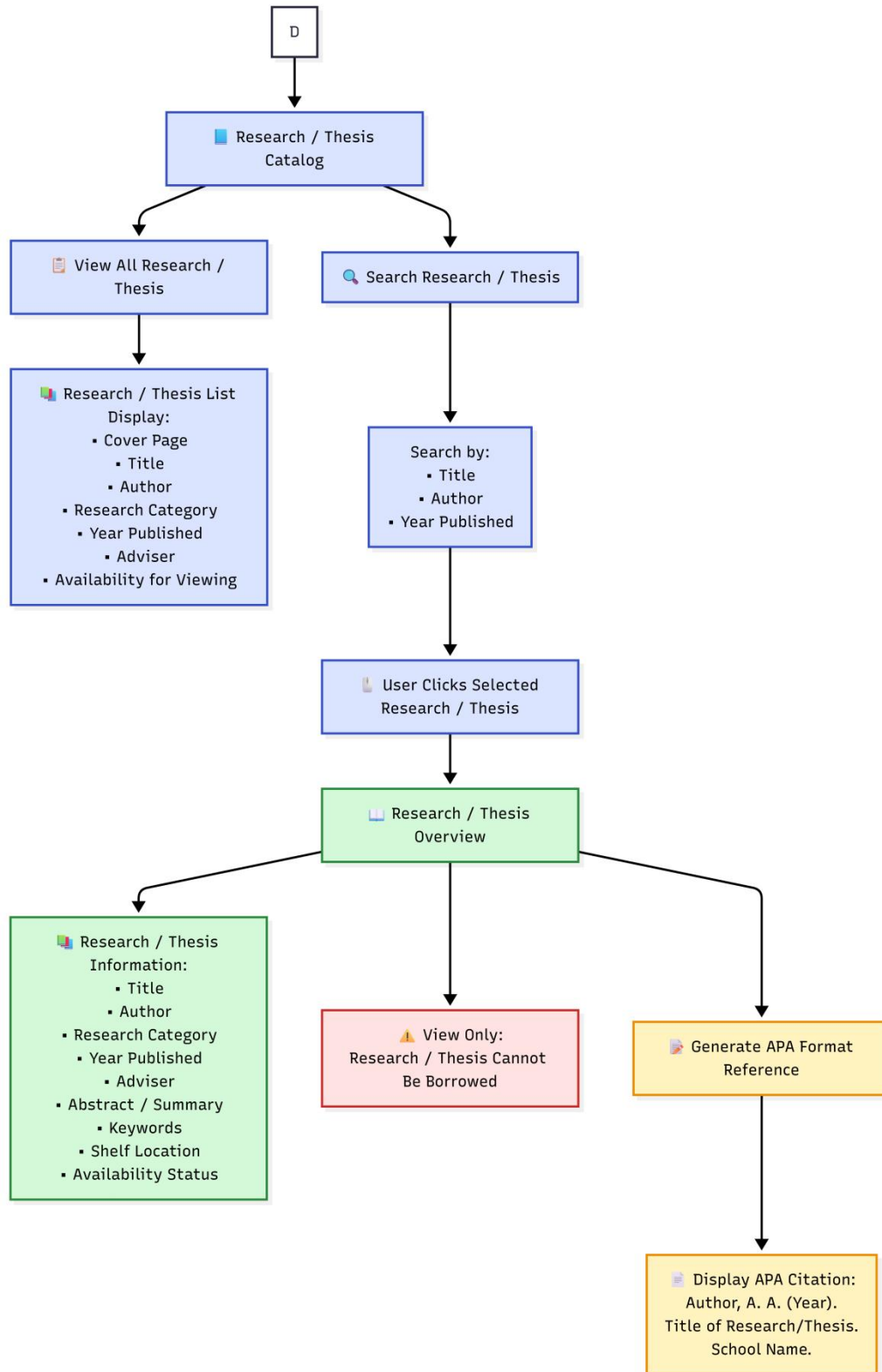


Figure 28. Proposed Research/Thesis Catalog Process (User Side)

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APPENDIX A. RESOURCE PERSONS

Engr. Sheena Joy S. Muyuela

Profession: Academic Head

Name of Institution: STI College Ormoc

Rona Mira B. Lucañas

Profession: IT Program Head / Capstone Project Coordinator

Name of Institution: STI College Ormoc

Elizabeth L. Dumaran

Profession: IT Instructor / Capstone Project Adviser

Name of Institution: STI College Ormoc

Judelyn Martinez

Profession: Librarian

Name of Institution: STI College Ormoc

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of
JOHN BAZTY C. CANTAY
BARANGAY TUGBONG, SITIO PUNAY KANANGA LEYTE
cantay242@gmail.com



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI College Ormoc
Vocational/Technical	2020-2023	KNHS Senior Highschool
High School	2016-2020	Kananga National Highschool
Elementary	2011-2016	Cacao Elementary Highschool

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
	N/A	

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member
2024	IT Club	Member
2023	IT Club	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
HTML	Entry	September 2025
CSS	Entry	September 2025
UI/UX Design	Intermediate	November 2025
Node.js	Entry	March 2026

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
September 2025	PayTaca Workshop
December 2024	National Service Training Program

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
KEITH GAUDI GERARD P. JUNTILLA
BRGY. LIBERTAD, ORMOC CITY, LEYTE
juntilla.keith@gmail.com
09602700364



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2025 - Current	STI College Ormoc
Vocational/Technical	2018 - 2021	Saint Peter's College of Ormoc
High School	2014 - 2018	Saint Peter's College of Ormoc
Elementary	2008 - 2014	Saint Peter's College of Ormoc

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
	N/A	

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Web Development (HTML & CSS)	Basic	January 2026
Android Studio (Mobile Dev)	Basic	January 2026
C, C#, Java Programming	Basic	August 2025

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
April 2026	Research Seminar: Bridging Academe & Industry through Research
November 2025	Cybersecurity: Digital First Responders - From Evidence to Ethics Seminar

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
BUENTHEO NATHANIEL A. NOVAL
BARANGAY PUNTA, ORMOC CITY LEYTE
ethannoal99@gmail.com



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI College Ormoc
Vocational/Technical	2020-2023	STI College Ormoc
High School	2016-2020	NOCNHS
Elementary	2011-2016	Ormoc City Central School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2022-2023	Graphics Designer	WELLIFE

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member
2024	IT Club	Member
2023	IT Club	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
HTML	Intermediate	January 2021
CSS / Tailwind CSS	Intermediate	January 2021
Javascript	Intermediate	January 2023
Frontend debugging	Intermediate	March 2024

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
September 2025	PayTaca Workshop
December 2024	National Service Training Program

Listed in reverse chronological order (most recent first).

APPENDIX C. TRANSMITTAL LETTER



May 4, 2026

Ms. Judelyn Martinez

Librarian

STI College – Ormoc

Subject: Request for Requirements Gathering and Workflow Consultation

Dear Ma'am,

Good day.

Following our initial proposal for the Automated Library Management System (LMS), we, the researchers, are now entering the critical phase of requirements gathering. To ensure that the final system effectively addresses the specific needs of our campus, we respectfully request a short window of your time today, May 4, 2026, to conduct a brief interview.

Our primary objectives for this visit are:

- To gain a deeper understanding of the library's day-to-day operations and current administrative workflows.
- To further discuss and understand the desired functionalities of the system's features to ensure they provide maximum utility to both staff and students.
- To identify opportunities where digital automation can best support the library's staff and improve service delivery to students.
- To gather feedback on proposed system functionalities and ensure they integrate seamlessly with the library's long-term goals.
- To discuss the specific data management and reporting requirements essential for efficient library oversight.

We understand that your schedule is busy and will ensure our consultation is concise and focused. Our goal is to collaborate closely with you to build a system that minimizes manual workload and enhances the library experience for all STIers.

Thank you for your continued support and guidance in this project. We look forward to your favorable response.

Sincerely yours,

Ethan Noval

John Bazty Cantay

Keith Gaudi Gerard Juntilla

APPENDIX D. PROJECT DOCUMENTATION

